

# Speech-to-Text Models for 1-Hour Audio (2026)

This document compares major AI speech-to-text (STT) services for analysing about 1 hour of audio, focusing on time-to-results, efficiency, and cost, and highlights which models are best suited for subtitles and captions. Prices are list rates as of early 2026 and rounded for simplicity; real-world bills depend on volume, regions, and add-ons.

## 1. Summary: Cost to Transcribe 1 Hour (Batch, Pre-recorded Audio)

Typical per-hour list pricing for batch transcription of 1 hour of pre-recorded audio: - OpenAI Whisper API / GPT-4o Transcribe Mini: about \$0.18–\$0.36 per hour depending on model (gpt-4o-mini-transcribe vs Whisper Large / gpt-4o-transcribe). - AssemblyAI Universal / Universal-3 Pro: around \$0.15–\$0.21 per hour for batch speech-to-text. - Deepgram Nova-3 (pre-recorded): roughly \$0.26–\$0.31 per hour, depending on tier and configuration. - Google Cloud Speech-to-Text (Standard): about \$0.96 per hour at the base tier, with volume discounts. - Amazon Transcribe (Standard): about \$1.44 per hour at Tier 1 (0–250k minutes/month). - Azure Speech-to-Text (batch / fast transcription): commonly in the \$0.18–\$0.36 per hour range for standard models. For pure transcription cost per hour of audio, OpenAI Whisper, AssemblyAI, and Deepgram are usually the most economical, while Google, AWS, and Azure are more expensive but may be preferable if you are already tightly integrated into those cloud ecosystems.

## 2. Time-to-Result for 1 Hour of Audio

Time-to-result depends on whether you use on-prem/Open Source or managed cloud APIs, and whether you run in batch or streaming mode: - Whisper (self-hosted): on a modern GPU, reported real-time factors (RTF) around 0.1–0.3x for large models mean 1 hour of audio can often be processed in roughly 6–18 minutes; on CPU, RTF of 1–3x means 1 hour of audio may take 1–3 hours. - Managed Whisper / AssemblyAI / Deepgram batch APIs: designed for near-real-time or faster-than-real-time processing; for 1 hour files, end-to-end turnaround is commonly in the range of a few minutes, depending on queueing, file size limits, and concurrency. - Google, AWS, and Azure batch APIs: also support faster-than-real-time processing for 1 hour clips, but overall latency can be higher because of queueing, region selection, or additional pipeline steps (storage upload, Cloud Functions, Lambdas, etc.). - Streaming use cases: when you stream a live 1 hour event, you get tokens back continuously; for live captions the relevant metric is per-utterance latency rather than total processing time. Leading APIs like Deepgram and AssemblyAI report p95 streaming latencies in the sub-300–400 ms range, while Google Chirp and others are slightly slower but still under a second for most workloads.

## 3. Efficiency: Accuracy and Resource Use

Efficiency here means accuracy (Word Error Rate, WER) per dollar and operational overhead: - Accuracy leaders: independent and vendor benchmarks typically show OpenAI Whisper and AssemblyAI Universal-3 Pro among the lowest WER across English and multilingual datasets, with Deepgram Nova-3 and Google Chirp close behind depending on audio type. - Noise and accents: Deepgram and AssemblyAI tend to be strong on noisy and accented speech; Whisper also performs well but may degrade somewhat more on heavily accented conversational audio compared to specially tuned commercial APIs. - Cloud provider trade-offs: Google, AWS, and Azure STT can be slightly less accurate than specialized providers for general audio, but are often good enough and attractive if you are already invested in the respective cloud (IAM, logging, monitoring, data residency). - Operational overhead: self-hosting Whisper or other open-source models gives you very low marginal per-hour costs but shifts the burden to your infra (GPU servers, scaling, monitoring). Managed APIs charge per minute but remove most ops work and simplify scaling to large numbers of hours.

## 4. Subtitle & Caption Workloads: What Matters

For subtitles and captions, you care about more than raw text: - Timestamp quality: you usually want at least phrase-level timestamps, ideally word-level timestamps for precise subtitle alignment and karaoke-style highlighting. Whisper (API and open source), AssemblyAI, Deepgram, Google, AWS, and Azure all provide timestamps; Whisper's verbose JSON and Deepgram/AssemblyAI metadata are popular in subtitle pipelines. - Punctuation and casing: modern APIs infer punctuation and sentence boundaries, which is critical for readability in subtitles. Whisper and the commercial APIs all handle this, though default styles can differ. - Diarization (who spoke when): important for multi-speaker videos, interviews, or panels.